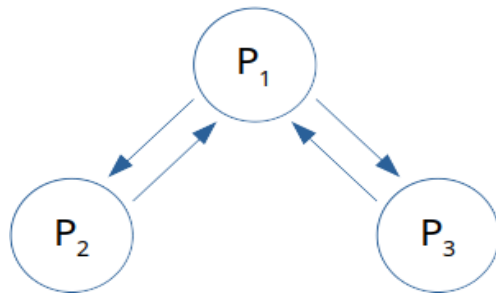


CMPE 382 Operating Systems, Spring 2025-2026

Homework 1-Multiprocess concept: fork () and pipe () system calls

Due Date: March 22, 2026, 23:59

In this homework, you are expected to write a C program, which demonstrates inter-process-communication-IPC. Starting program is the parent process. It creates two other (child) processes and establish a multi-directional communication as shown in the figure below:



The parent process P1 reads an integer N and creates two files, each contains N positive random integers in the range of 0-XXX, Where XXX is the last 3 digits of your ID number plus 100. Let file names be File1, and File2. Then, the parent shares N with the child processes over the pipes. In return it receives the number of prime numbers, and abundant numbers in the respective files from the child processes and shares them with the other child processes to see which one is the winner. The process having the highest count is the winner.

To be more specific, each child process receives N from the parent, finds the count of prime numbers, and abundant numbers (PN, AN) in respective files File1 (PN), File2(AN) and shares it with the parent. Each child now receives the count of numbers of the other child process from the parent. Each will now know which child process has the higher count and prints the following message.

"I am Child process Px: The winner is child process Py", where x is the number of the child process writing the message, y is the winner process.

The parent processor prints the following message on the completion of the child processes:

"The number of positive integers in each file: xxxx

The number of prime numbers in File1: xxxx

The number of abundant numbers in File2: xxxx"

Further Notes

1. No team working is allowed.
2. You may study from other sources, but you are strictly forbidden to copy from them.
3. Use C programming language. It must compile in GNU/Linux and execute with no errors.
4. Name your program as CMPE382-HW1_YourInitials-YourID such as "CMPE382-HW1_MB-21326452825"
5. Upload the source program and output on LMS.
6. For code generation, using any GPT-based AI tools isn't allowed. You have to write the comments to your code. Your code will be evaluated through AI plagiarism.